

A group of students were given water from two reservoirs and asked to determine water hardness using flame atomic absorbance (AA) spectroscopy. Water hardness can be found by relating the concentration of calcium and magnesium ions to the concentration of carbonate species in the sample using Equation 1.

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Equation 1

To correlate the concentration of metal ions to absorbance, the students made standard solutions of MgO (MW = 40.31 g/mol) and CaO (MW = 56.08 g/mol) by diluting small portions of 0.001 M MgO and 0.01 M CaO stock solutions with 2% HNO₃ into 100 mL volumetric flasks. The calibration curves generated from the standard solutions are shown in Figures 1 and 2.

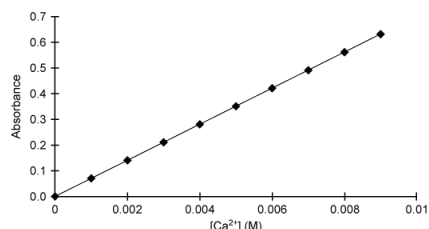


Figure 1 Calibration curve from the absorption of the Ca²⁺ solution standards

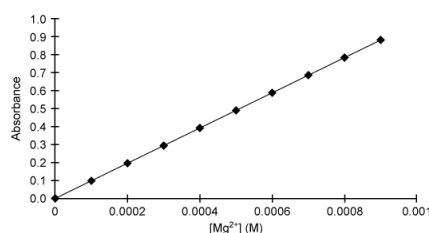


Figure 2 Calibration curve from the absorption of the Mg²⁺ solution standards

Lanthanum(III) chloride (MM = 245.25) can be used to eliminate chemical interference in a sample by competitively binding to PO₄³⁻, releasing calcium to be atomized in the flame for detection.

The students prepared unknown Samples 1 and 2 by mixing 25 mL of reservoir water, 16 mL of 2% HNO₃, and 9 mL of 0.1 M LaCl₃ for a total sample volume of 50 mL each. The samples were each tested three times using flame AA spectroscopy, yielding the results reported in Table 1.

Table 1 AA Spectroscopy Results for Reservoir Water Samples

Source	Calcium (average absorbance)	Magnesium (average absorbance)
Sample 1	0.35	0.49
Sample 2	0.20	0.30

What is the molar concentration of Cl⁻ ions in a solution prepared by dissolving 0.1 moles of lanthanum(III) chloride in 1 L of water?

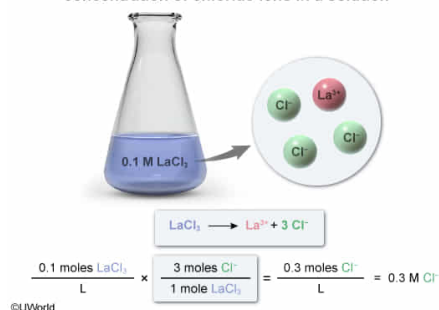
- ☐ A. 0.03 M [8%]
☐ B. 0.1 M [14%]
☒ C. 0.3 M [76%]
☐ D. 30 M [1%]

Correct

75% answered correctly

Explanation:

Calculating the molar concentration of chloride ions in a solution



The **mole ratio** is the whole number proportion of each chemical species in a balanced chemical equation or molecular formula. If the amount of a chemical substance is given in something other than moles (eg, grams), then this amount must be converted to moles first to relate the various species proportions.

Molarity is defined as moles of solute per liter of solution. Because molarity expresses concentration in terms of moles, the mole ratio of the solute ions can be multiplied by the concentration of LaCl_3 in solution to find the concentration of individual ions in solution.

The concentration of LaCl_3 is 0.1 M and, based on the molecular formula, there are 3 moles of Cl^- ions in every 1 mole of LaCl_3 . The concentration of LaCl_3 can be related to the concentration of Cl^- ions by using this mole ratio as a conversion factor:

$$\frac{0.1 \text{ mol } \text{LaCl}_3}{\text{L}} \times \frac{3 \text{ mol } \text{Cl}^-}{1 \text{ mol } \text{LaCl}_3} = \frac{0.3 \text{ mol } \text{Cl}^-}{\text{L}} = 0.3 \text{ M } \text{Cl}^-$$

Therefore, the concentration of Cl^- ions must be 0.3 M.

(Choice A) 0.03 M is obtained by dividing rather than multiplying the concentration of LaCl_3 by 3 moles Cl^- . This operation does not cancel out the moles of LaCl_3 , and therefore provides **incorrect units**.

(Choice B) 0.1 M is the concentration of LaCl_3 . The concentration of Cl^- must be larger than the concentration of LaCl_3 because there are 3 Cl^- ions for every La^{3+} .

(Choice D) 30 M is obtained if the mole ratio of Cl^- to LaCl_3 is divided by (instead of multiplied by) the concentration of LaCl_3 . This will not cancel out the moles of LaCl_3 , resulting in an answer with **incorrect units**.

Educational objective:

The mole ratio is the whole number proportion of each chemical species in a balanced chemical equation. Molarity is the number of moles of solute per liter of solution. Because molarity expresses concentration in terms of moles, a mole ratio can be used to relate the concentration (molarity) of one species to the concentration of another species in solution.

Time spent: 0 sec
Subject:
 General Chemistry

QID: 401552
System:
 Atomic Theory and Chemical Composition

A group of students were given water from two reservoirs and asked to determine water hardness using flame atomic absorbance (AA) spectroscopy. Water hardness can be found by relating the concentration of calcium and magnesium ions to the concentration of carbonate species in the sample using Equation 1.

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To correlate the concentration of metal ions to absorbance, the students made standard solutions of MgO (MW = 40.31 g/mol) and CaO (MW = 56.08 g/mol) by diluting small portions of 0.001 M MgO and 0.01 M CaO stock solutions with 2% HNO₃ into 100 mL volumetric flasks. The calibration curves generated from the standard solutions are shown in Figures 1 and 2.

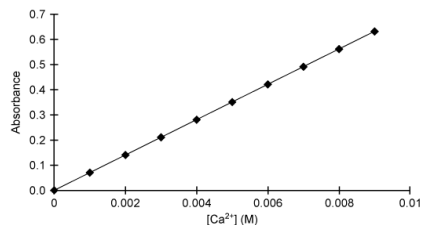


Figure 1 Calibration curve from the absorption of the Ca²⁺ solution standards

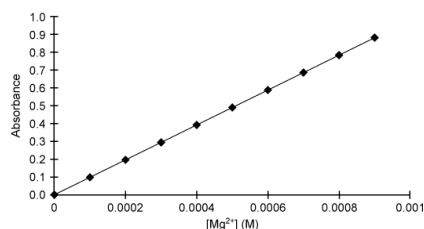


Figure 2 Calibration curve from the absorption of the Mg²⁺ solution standards

Lanthanum(III) chloride (MM = 245.25) can be used to eliminate chemical interference in a sample by competitively binding to PO₄³⁻, releasing calcium to be atomized in the flame for detection.

The students prepared unknown Samples 1 and 2 by mixing 25 mL of reservoir water, 16 mL of 2% HNO₃, and 9 mL of 0.1 M LaCl₃ for a total sample volume of 50 mL each. The samples were each tested three times using flame AA spectroscopy, yielding the results reported in Table 1.

Table 1 AA Spectroscopy Results for Reservoir Water Samples

Source	Calcium (average absorbance)	Magnesium (average absorbance)
Sample 1	0.35	0.49
Sample 2	0.20	0.30

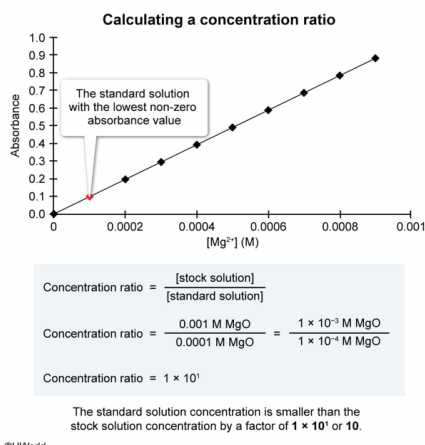
The passage states that the MgO standard solutions were prepared for analysis by diluting small portions of the 0.001 M MgO stock solution. Compared to the concentration of the MgO standard solution with the lowest nonzero absorbance, the concentration of the stock solution differs by what factor?

- ☐ A. 1×10^{-7} [4%]
☐ B. 1×10^{-1} [30%]
☒ C. 1×10^1 [49%]
☐ D. 1×10^2 [15%]

Correct

50% answered correctly

Explanation:



To make **standard solutions** of a particular solute, a certain volume (or aliquot) of **stock solution** is added to a flask and **diluted** with solvent to achieve the desired solute concentration. The **amount of solute in the original** aliquot of stock solution and the amount of solute in the diluted standard solution are the same; the only difference is the concentration. A **ratio of the concentrations** can be used to determine how much larger or smaller the concentration of one solution is compared to the other.

The standard solution with the lowest nonzero absorbance value is the solution with the lowest concentration. According to the Mg^{2+} calibration curve shown in Figure 2, this would be the 0.0001 M solution of Mg^{2+} . The factor by which the concentration of the stock solution differs from concentration of the standard solution is determined by the concentration ratio:

$$\text{Concentration ratio} = \frac{[\text{stock solution}]}{[\text{standard solution}]}$$

$$\text{Concentration ratio} = \frac{1 \times 10^{-3} \text{ M}}{1 \times 10^{-4} \text{ M}} = 1 \times 10^1$$

This ratio is known as the **dilution factor** and can also be determined by taking the ratio of the volume of the diluted solution to the volume of original solution used.

(Choice A) An incorrect factor of 1×10^{-7} is obtained by multiplying the concentrations (ie, **adding the exponents**: $-3 + -4 = -7$) rather than dividing (ie, subtracting the exponents).

(Choice B) A factor of 1×10^{-1} is obtained by incorrectly setting up the concentration ratio as $\frac{[\text{standard solution}]}{[\text{stock solution}]}$ rather than $\frac{[\text{stock solution}]}{[\text{standard solution}]}$.

(Choice D) An incorrect factor of 1×10^2 is obtained by correctly setting up the molar ratio but incorrectly determining the standard concentration to be $1 \times 10^{-5} \text{ M}$ rather than $1 \times 10^{-4} \text{ M}$.

Educational objective:

A standard solution can be made by diluting a particular amount of a stock solution. Because the number of solute particles stays the same, the ratio of concentrations can be calculated to determine the factor by which the concentration of the standard solution is diluted compared to the stock solution.

Time spent: 0 sec
Subject:
General Chemistry

QID: 401553
System:
Atomic Theory and Chemical Composition

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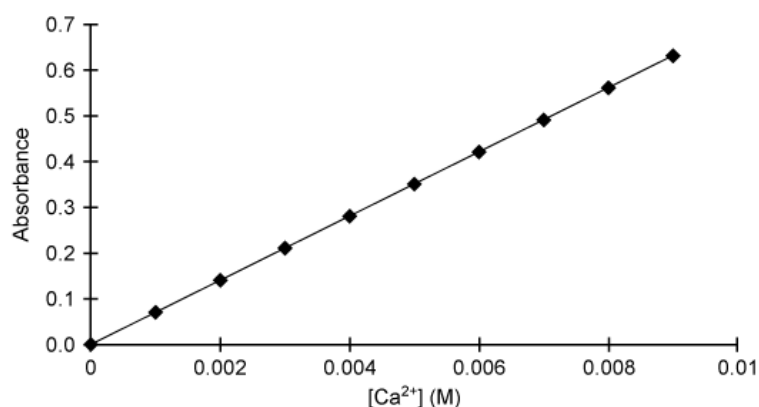
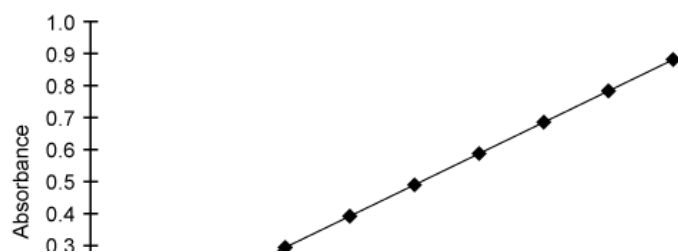


Figure 1 Calibration curve from the absorption of the Ca^{2+} solution standards



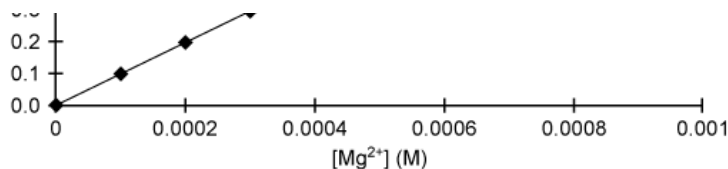


Figure 2 Calibration curve from the absorption of the Mg^{2+} solution standards

Lanthanum(III) chloride (MM = 245.25) can be used to eliminate chemical interference in a sample by competitively binding to PO_4^{3-} , releasing calcium to be atomized in the flame for detection.

The students prepared unknown Samples 1 and 2 by mixing 25 mL of reservoir water, 16 mL of 2% HNO_3 , and 9 mL of 0.1 M LaCl_3 for a total sample volume of 50 mL each. The samples were each tested three times using flame AA spectroscopy, yielding the results reported in Table 1.

Table 1 AA Spectroscopy Results for Reservoir Water Samples

Source	Calcium (average absorbance)	Magnesium (average absorbance)
Sample 1	0.35	0.49
Sample 2	0.20	0.30

Approximately how many La^{3+} ions are present in each of the unknown samples?

- ☐ A. 1×10^{-27} [4%]
- ✓ ☒ B. 5×10^{20} [59%]
- ☐ C. 9×10^{20} [19%]
- ☐ D. 6×10^{22} [16%]

Correct

59% answered correctly

Explanation:

Determining the number of lanthanum ions in a sample

$$\begin{array}{ccccccc} & & \text{LaCl}_3 \longrightarrow \text{La}^{3+} + 3\text{Cl}^- & & & & \\ \text{Given volume} & & \text{Given concentration} & & \text{Mole ratio} & & \text{Moles of LaCl}_3 \text{ in the} \\ \text{of LaCl}_3 & & \text{of LaCl}_3 & & & & \text{unknown sample} \\ 9 \text{ mL} \times \frac{1 \text{ L}}{1,000 \text{ mL}} \times \frac{0.1 \text{ moles LaCl}_3}{\text{L}} \times \frac{1 \text{ mole La}^{3+}}{1 \text{ mole LaCl}_3} = 9 \times 10^{-4} \text{ moles of La}^{3+} \\ \\ & & \text{Avogadro's number} & & & & \text{Resulting number of} \\ & & \text{relates moles to ions} & & & & \text{La}^{3+} \text{ ions in the} \\ & & & & & & \text{unknown samples} \\ 9 \times 10^{-4} \text{ moles of La}^{3+} \times \frac{6.02 \times 10^{23} \text{ ions}}{\text{mole}} = 54 \times 10^{19} = 5.4 \times 10^{20} \text{ ions of La}^{3+} \end{array}$$

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Even though chemical compounds are made of atoms, it is impractical to use the number of atoms to relate one compound to another in a chemical reaction because the actual number of atoms is very large. Instead, chemists use the *mole* to keep track of atoms. A **mole** is defined as the amount of a substance that contains as many particles, atoms, or ions as the number of atoms in 12 grams of ^{12}C . Experimental studies have demonstrated that there are 6.02×10^{23} atoms in 12 grams of ^{12}C , a value known as **Avogadro's number**. Moles of a substance can be **converted to ions** using [Avogadro's number](#).

The passage states 9 mL of 0.1 M LaCl_3 is added to each unknown sample mixture. Although this mixture is ultimately diluted to a total volume of 50 mL, the 9 mL of 0.1 M LaCl_3 contains all the La^{3+} ions present in the mixture. To determine the number of moles of LaCl_3 present in the final mixture, the added volume of the LaCl_3 stock solution must first be converted to liters and then multiplied by its [molar concentration](#):

$$9 \text{ mL} \times \frac{1 \text{ L}}{1,000 \text{ mL}} \times \frac{0.1 \text{ mol LaCl}_3}{\text{L}} = 9 \times 10^{-4} \text{ mol LaCl}_3$$

Based on the compound formula, the moles of La^{3+} are equivalent to the number of moles of LaCl_3 because the **mole ratio** of $\text{La}^{3+}:\text{LaCl}_3$ is 1:1.

$$9 \times 10^{-4} \text{ mol } \text{LaCl}_3 \times \frac{1 \text{ mol } \text{La}^{3+}}{1 \text{ mol } \text{LaCl}_3} = 9 \times 10^{-4} \text{ mol } \text{La}^{3+}$$

Finally, **multiplying** the moles of La^{3+} by Avogadro's number and reporting the value with the correct **scientific notation** yields the number of La^{3+} ions :

$$9 \times 10^{-4} \text{ mol } \text{La}^{3+} \times \frac{6.02 \times 10^{23} \text{ La}^{3+} \text{ ions}}{1 \text{ mol } \text{La}^{3+}} = 5 \times 10^{20} \text{ La}^{3+} \text{ ions}$$

Therefore, approximately 5×10^{20} La^{3+} ions are present in each unknown sample.

(Choice A) 1×10^{-27} ions would be obtained if moles of La^{3+} were *divided by*, instead of multiplied by, Avogadro's number.

(Choice C) 9×10^{20} ions would be obtained if 1×10^{23} was used for Avogadro's number rather than 6.02×10^{23} .

(Choice D) 6×10^{22} ions would be obtained if the volume of LaCl_3 used and its conversion from milliliters to liters is not considered in the calculation. In such a case, the number of moles would not be determined, and the concentration of LaCl_3 would simply be multiplied by Avogadro's number.

Educational objective:

A mole is the amount of a substance that contains as many particles, ions, or atoms as the number of atoms in 12 grams of ^{12}C . Moles of a substance can be converted to ions (or particles,

molecules, atoms) using Avogadro's number: 6.02×10^{23} .

Time spent:0 sec

Subject:
General Chemistry

QID:401554

System:
Atomic Theory and Chemical Composition

A group of students were given water from two reservoirs and asked to determine water hardness using flame atomic absorbance (AA) spectroscopy. Water hardness can be found by relating the concentration of calcium and magnesium ions to the concentration of carbonate species in the sample using Equation 1.

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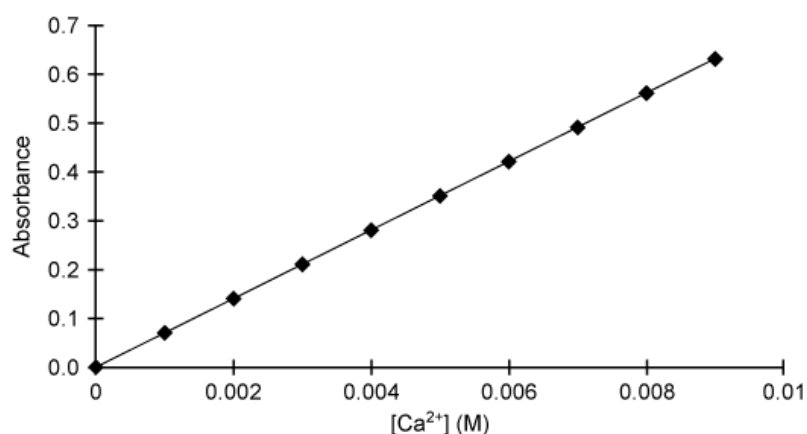


Figure 1 Calibration curve from the absorption of the Ca²⁺ solution standards

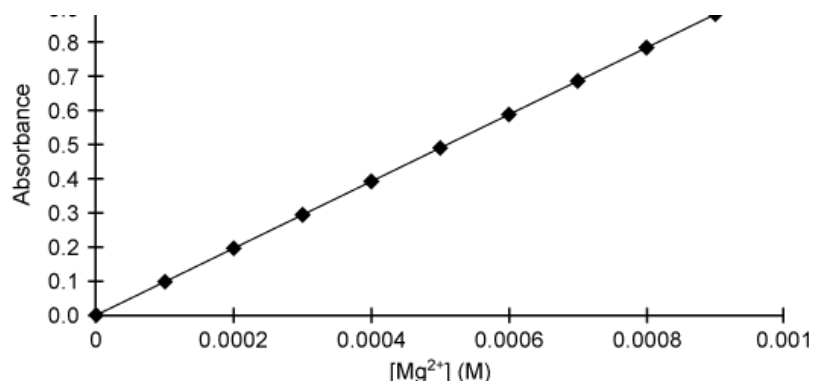


Figure 2 Calibration curve from the absorption of the Mg^{2+} solution standards

Lanthanum(III) chloride ($MM = 245.25$) can be used to eliminate chemical interference in a sample by competitively binding to PO_4^{3-} , releasing calcium to be atomized in the flame for detection.

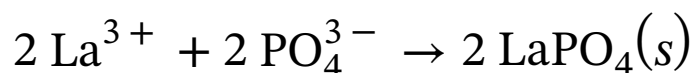
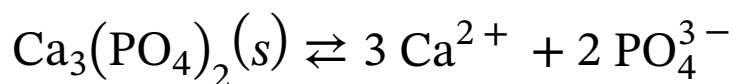
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M $LaCl_3$ for a total sample volume of 50 mL each. The samples were each tested three times using flame AA spectroscopy, yielding the results reported in Table 1.

Table 1 AA Spectroscopy Results for Reservoir Water Samples

Source	Calcium (average absorbance)	Magnesium (average absorbance)
Sample 1	0.35	0.49
Sample 2	0.20	0.30

When $LaCl_3$ is added in excess to a sample of reservoir water, the following reactions take place:



If 2×10^{-4} moles of $\text{Ca}_3(\text{PO}_4)_2$ react, the maximum number of grams of LaPO_4 (MW = 234 g/mol) that can be produced is:

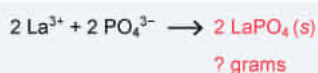
- ☐ A. 2.3×10^{-2} g [6%]
- ☐ B. 3.1×10^{-2} g [5%]
- ☐ C. 4.7×10^{-2} g [15%]
- ✓ ☒ D. 9.4×10^{-2} g [72%]

Correct

71% answered correctly

Explanation:

Determining the mass of LaPO_4 produced during a reaction



The coefficients
relate moles of
reactant to moles
of product.

Molar mass
converts moles
to grams.

Resulting grams
of LaPO_4

$$2 \times 10^{-4} \text{ mol Ca}_3(\text{PO}_4)_2 \times \frac{2 \text{ mol LaPO}_4}{1 \text{ mol Ca}_3(\text{PO}_4)_2} \times \frac{234 \text{ g LaPO}_4}{\text{mol}} = 9.4 \times 10^{-2} \text{ g LaPO}_4$$

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The **stoichiometric coefficients** in a **balanced chemical reaction** provide the **mole ratios** of all species involved in the reaction. These ratios are useful to convert the quantity of moles of one species to the quantity of moles of another species in the reaction. **Moles** can then be **converted to grams** using the **molar mass**.

The two given reactions are connected by the PO_4^{3-} ion such that $\text{Ca}_3(\text{PO}_4)_2$ is the reactant and LaPO_4 is the product of the overall reaction. Because LaCl_3 is added in excess, $\text{Ca}_3(\text{PO}_4)_2$ is the **limiting reactant**. Based on the balanced chemical reaction, 2 moles of LaPO_4 are produced for every 1 mole of $\text{Ca}_3(\text{PO}_4)_2$ that reacts. The maximum number of moles of LaPO_4 possible is calculated by multiplying the given number of moles of $\text{Ca}_3(\text{PO}_4)_2$ by the molar ratio:

$$2 \times 10^{-4} \text{ mol } \text{Ca}_3(\text{PO}_4)_2 \times \frac{2 \text{ mol } \text{LaPO}_4}{1 \text{ mol } \text{Ca}_3(\text{PO}_4)_2} = 4 \times 10^{-4} \text{ mol } \text{LaPO}_4$$

The number of moles of LaPO_4 is then multiplied by its molar mass to give the number of grams of LaPO_4 :

$$4 \times 10^{-4} \text{ mol } \text{LaPO}_4 \times \frac{234 \text{ g } \text{LaPO}_4}{1 \text{ mol } \text{LaPO}_4} = 9.4 \times 10^{-2} \text{ g } \text{LaPO}_4$$

(Choice A) A mass of $2.3 \times 10^{-2} \text{ g } \text{LaPO}_4$ would be obtained by using an inverted molar ratio of 1:2 instead of 2:1.

(Choice B) If Ca^{2+} were used as the reactant rather than $\text{Ca}_3(\text{PO}_4)_2$, the molar ratio would be 2:3, giving $3.1 \times 10^{-2} \text{ g } \text{LaPO}_4$.

(Choice C) If the stoichiometric ratios in the balanced chemical equation were not observed and a 1:1 molar ratio between $\text{Ca}_3(\text{PO}_4)_2$ and LaPO_4 were used, a mass of $4.7 \times$

10^{-2} g LaPO_4 would be obtained.

Educational objective:

The mass of a product can be calculated given a particular amount of reactant by first calculating the number of moles of product formed using a reactant-to-product mole ratio derived from the balanced chemical equation. The moles of product can then be converted to grams using the molar mass of the product.

Time spent:0 sec

Subject:
General Chemistry

QID:401555

System:
Atomic Theory and Chemical
Composition

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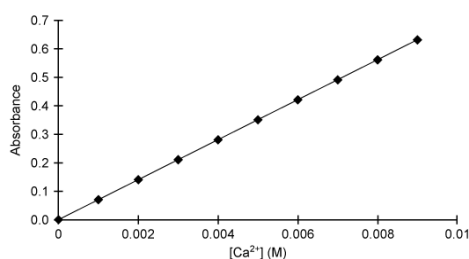


Figure 1 Calibration curve from the absorption of the Ca²⁺ solution standards

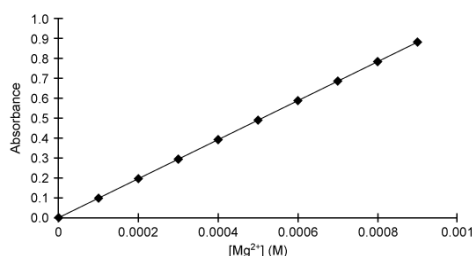


Figure 2 Calibration curve from the absorption of the Mg²⁺ solution standards

Lanthanum(III) chloride (MM = 245.25) can be used to eliminate chemical interference in a sample by competitively binding to PO₄³⁻, releasing calcium to be atomized in the flame for detection.

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Source	Calcium (average absorbance)	Magnesium (average absorbance)
Sample 1	0.35	0.49
Sample 2	0.20	0.30

Based on the information in the passage, what is the mass of calcium (MW = 40.1 g/mol) in the Sample 1 mixture?

- ☐ A. 0.003 g [10%]
☐ B. 0.005 g [23%]
☒ C. 0.01 g [61%]
☐ D. 10 g [5%]

Correct

61% answered correctly

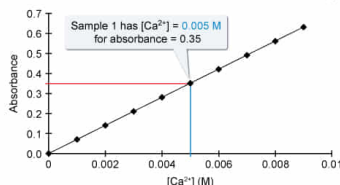
Explanation:

Calculating the mass of calcium in a sample of reservoir water

Step 1: Determine absorbance of Ca^{2+} from data table

Source	Calcium (average absorbance)	Magnesium (average absorbance)
Sample 1	0.35	0.49
Sample 2	0.20	0.30

Step 2: Determine concentration of Ca^{2+} from calibration curve using absorbance



Step 3: Use total volume, concentration, and molar mass to calculate mass of Ca^{2+}

Total volume	Unit conversion	Concentration	Molar mass	Mass
50 mL	$\frac{1 \text{ L}}{1000 \text{ mL}}$	$\frac{0.005 \text{ mol Ca}^{2+}}{1 \text{ L}}$	$\frac{40.1 \text{ g Ca}^{2+}}{1 \text{ mol Ca}^{2+}}$	= 0.01 g Ca^{2+}

To find the mass of a component of an unknown sample, the unknown quantity needs to be related to a known quantity. **Standard solutions** are solutions of a known quantity (concentration) of the component being studied (**analyte**) in the unknown samples. Standard solutions are used to make a **calibration curve**, which relates the amount of the analyte (x-axis) to the instrumental detection (y-axis).

Table 1 reports that the absorbance of Ca^{2+} detected in Sample 1 is 0.35. The calibration curve in Figure 1 provides absorbance values for known **molar concentrations** of Ca^{2+} , or $[\text{Ca}^{2+}]$, in the standard solutions. Based on Figure 1, the concentration of Ca^{2+} in Sample 1 is approximately 0.005 M at an absorbance of 0.35.

The number of moles of Ca^{2+} is calculated by multiplying the *total* volume of the sample mixture (50 mL), converted from milliliters to liters so the units will cancel, by the concentration of Ca^{2+} :

$$50 \text{ mL} \times \frac{1 \text{ L}}{1,000 \text{ mL}} \times \frac{0.005 \text{ mol Ca}^{2+}}{1 \text{ L}} = 0.00025 \text{ mol Ca}^{2+}$$

Multiplying the **moles** of Ca^{2+} by its **molar mass** gives the **mass** of Ca^{2+} in Sample 1:

$$0.00025 \text{ mol Ca}^{2+} \times \frac{40.1 \text{ g Ca}^{2+}}{\text{mol Ca}^{2+}} = 0.01 \text{ g Ca}^{2+}$$

(Choice A) Although 0.003 g is calculated correctly, this result uses the absorbance of Sample 2 (0.19) rather than that of Sample 1 (0.35), and applies the incorrect value for $[\text{Ca}^{2+}]$ in the calculation.

(Choice B) If the volume of only the reservoir water (25 mL) was used instead of the *total* solution volume (50 mL), the calculated mass of Ca^{2+} would be 0.005 g.

(Choice D) If $[\text{Ca}^{2+}]$ was multiplied by 50 mL instead of 0.050 L, the answer would be 10 g.

Educational objective:

The analyte concentration in an unknown sample can be determined by comparing its absorbance to a calibration curve prepared using the measured absorbances of standard solutions with known concentrations. The determined concentration may then be used to calculate the mass of the analyte present in the sample based on the solution volume, relevant mole ratios, and the molar mass of the analyte.